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V300 ZEFIRO

A New Sense of Speed

Train Protection- VF.CONCEPT SFRS

Responsible Division:	Responsible Unit:	Document Type:	Distribution Status:	Document State:

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Identity Number:
EC01P008999B

Effective Date:	Revision:	Language:
27/01/2020	04	En

Revisionlog

Revision	Date	Changes	Reason
00	02/02/2012	Issue for PDM according to DOORS ECO1P008999B_SFRS_ATP Baseline 1.0.doc	
01.1	03/02/2012	Updated §3.1.2.6 to include also SCMT Electric brake command Added §3.1.2.8 Issue for PDM according to DOORS ECO1P008999B_SFRS_ATP Baseline 1.1.doc	
02.1	26/11/2012	SIL level of requirement 2F_09.02.Zefiro-Europe.CONCEPT.317 and. 384 downgraded to SIL0 Typo corrected and TBD removed in table Traceability matrix between SFRS and RD3 §5	
03.1	25/03/2013	§3.1.2.7 modified to avoid remote DIS traction cut off command. §3.1.2.8.1.1 modified to highlight that ATP will command the MCB only during phase and voltage transition) §3.1.3 updated to include reference to RD 8~~RD 8~~ and RD 9~~RD 9~~ for data to be recorded. Updated to have avaria DIS lamp on in case of DIS CB trip §5 updated reference documentation	
04.1	27/01/2020	Issue for alignment with Train Protection-VF.CONCEPT, baseline 2.2 as built	

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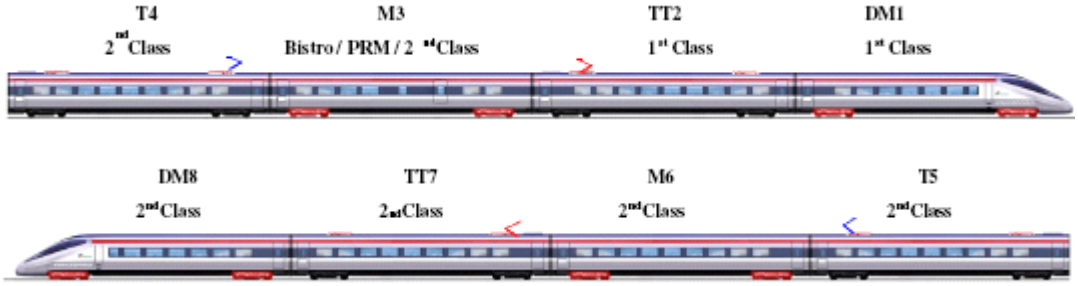
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1 Introduction

This document is a function specification for the control of the *ATP* system for V300ZEFIRO for customer Trenitalia.

1.1 Train Configuration

Each V300ZEFIRO train consists of eight cars in a fixed formation including a driver's cab at each end:



- DM = End driving motor car with driver's cab
- TT = Intermediate trailer car with transformer
- M = Intermediate driving motor car
- T = Intermediate trailer car

1.2 Definitions and abbreviations

ATESS NG+	Safety Static Event Temporal Acquisition New Generation
CD	Circuit Diagram
CEA	Commutatore Esclusione ATP (ATP Exclusion Switch)
CETD	Commutatore Escusione Tagio trazione DIS (DIS traction cut off exclusion switch)
DIS	Driver Information system
DVD	Driver Vigilance Device
EB	Emergency Brake
ETCS	European Train Control System
EVC	European Vital computer
EVENT	Diagnostics that will be stored in the Event database within TCMS SW.
EVIG	Esclusione Vigilante (DVD exclusion switch)
FS	Full supervision
HW HMI	Hardware Human Machine Interface, e.g. buttons, switches and indication lamps.
IDU	Interface Display Unit
JRU	Juridical Recorder Unit
MMI	Man Machine Interface
NSTOP	Non stopping area
RSC	Ripetizione Segnali Continua (italian National ATP)
SCMT	Sistema Controllo Marcia Treno (italian National ATP)
SR	Staff Responsible
TCMS	Train Control and Management System
TN	Tratto Neutro (Neutral Section)
TT	Taglio Trazione (Traction Cut off)

2 System Overview

2.1 System Configuration

The following figure shows the architecture of the ATP system for the “base” version:

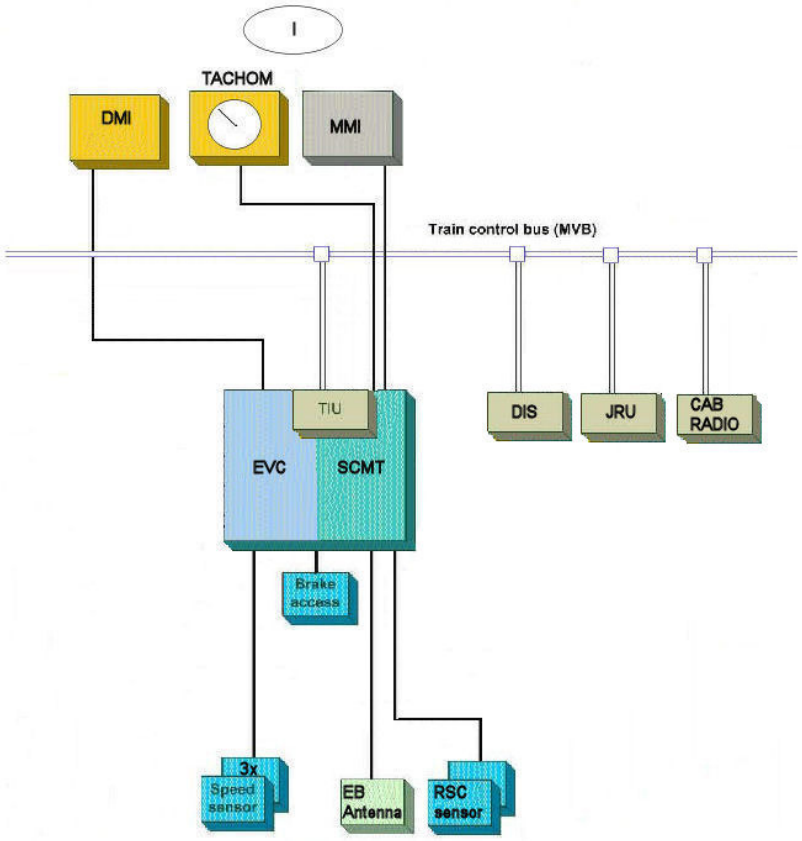
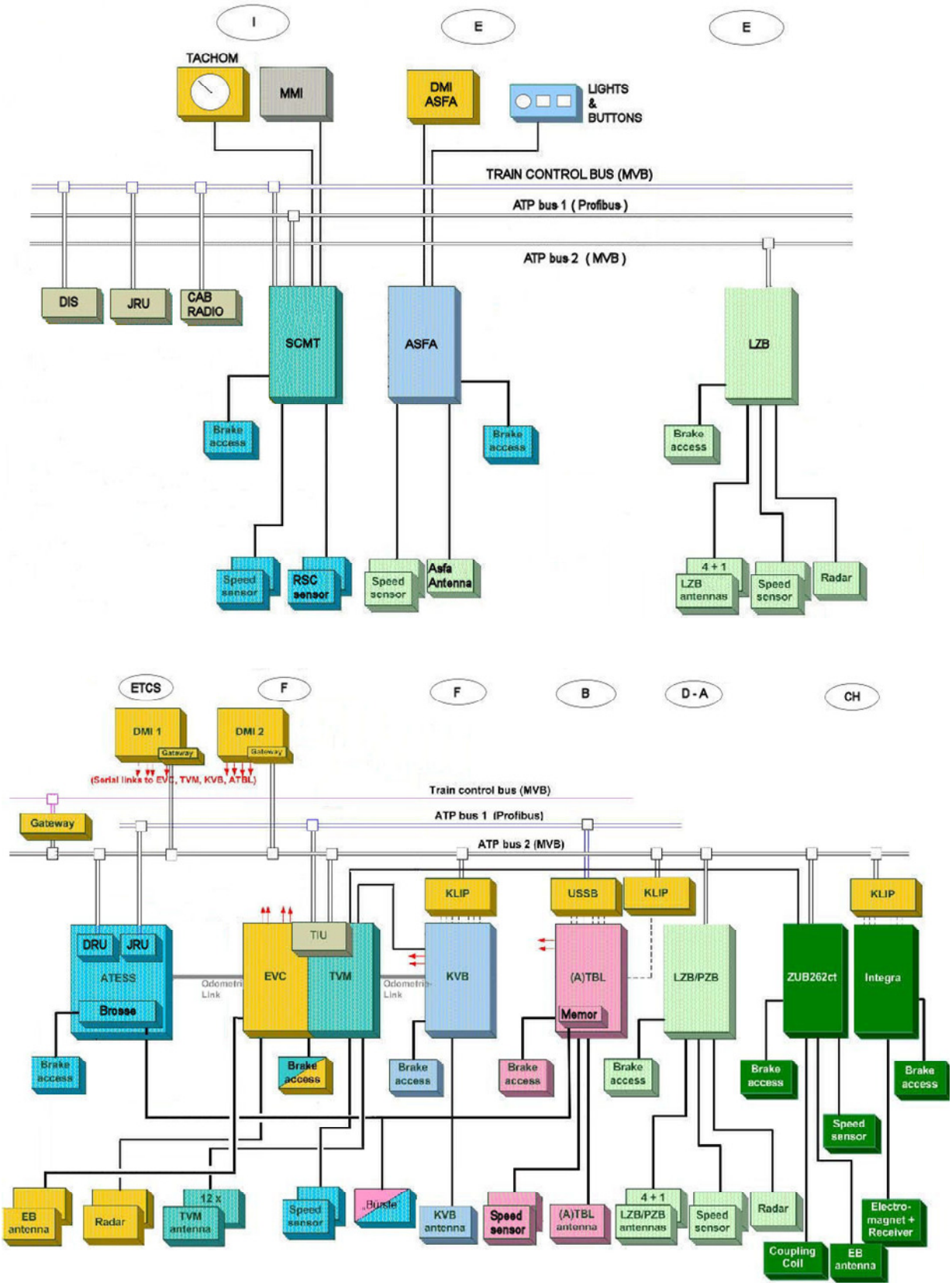


Figure 1: “base” configuration

The following figures show the architecture of the ATP system for the “corridors” version:


I
E
E
TACHOM
MMI
DMI ASFA
LIGHTS & BUTTONS
TRAIN CONTROL BUS (MVB)
ATP bus 1 (Profibus)
ATP bus 2 (MVB)
DIS
JRU
CAB RADIO
SCMT
ASFA
LZB
Brake access
Brake access
Brake access
Speed sensor
RSC sensor
Speed sensor
Asfa Antenna
4 + 1 LZB antennas
Speed sensor
Radar
ETCS
F
F
B
D - A
CH
DMI 1
DMI 2
Gateway
(Serial links to EVC, TVM, KVB, ATBL)
Train control bus (MVB)
ATP bus 1 (Profibus)
ATP bus 2 (MVB)
Gateway
KLIP
USSB
KLIP
KLIP
DRU
JRU
ATESS
Brosse
TIU
EVC
TVM
KVB
(A)TBL
Memor
LZB/PZB
ZUB262ct
Integra
Odometric Link
Odometric Link
Brake access
Brake access
Brake access
Brake access
Brake access
Brake access
Brake access
Speed sensor
EB antenna
Radar
12 x TVM antenna
Speed sensor
„Buste"
KVB antenna
Speed sensor
(A)TBL antenna
4 + 1 LZB/PZB antennas
Speed sensor
Radar
Electro-magnet + Receiver
Coupling Coil
EB antenna

The following table summarize the functions of the several equipment:

ATP functions	Equipment	Notes
ETCS Italia	ETCS/SCMT rack (ALA)	ETCS version SRS 2.3.0.d (subset 108 v1.2.0)
ETCS “corridoi”	ETCS/TVM rack (DIVA)	ETCS version SRS 2.3.0.d (subset 108 v1.2.0)
SCMT “Italy”	Integrated with ETCS	
SCMT “corridors”	SCMT rack	
TVM	Integrated with ETCS	
KVB	STM KVB FFFIS	
RPS	ATESS NG+	
LZB/PZB	LZB80E	
ATB	ATBL/USSB	
TBL 1/2	ATBL/USSB	
MEMOR	ATBL/USSB	
ZUB	ZUB262ct	

3 Functional requirements

3.1 Train Protection

3.1.1 Driver´s Safety Device (DSD)

The vigilance device for the “Base” configuration is integrated into the SCMT and therefore the DVD controls (pushbuttons on the traction lever and pedal) are managed by the SCMT control unit.
The vigilance device for the “corridors” has to be implemented in an additional control unit (ATESS NG+) and therefore there will be specific requirement for this configuration.

3.1.1.1 Driver´s Vigilance Device (DVD)

3.1.1.1.1 Function Design

The vigilance device monitor the presence and vigilance of the driver through the activation of the controls in the cabin.
In case the driver vigilance or presence is not confirmed the DVD switch on an optical warning, then acoustical warning and in the end it applies the emergency brake. The event is then recorded by the train recorder and a message is sent to the ground by the cab radio. The SCMT has two digital input to monitor the driver vigilance, the “primary chain” and the “secondary chain”.

Safety dvd control devices (pedal and button on traction lever) are connected to the primary chain. The secondary chain is connected to a TCMS output: the TCMS will monitor driver actions on several control (e.g IDU acknowled, IR command, horn command ecc…) and then will inform the SCMT of the driver vigilance. Refer to RD 10 and to the following picture.

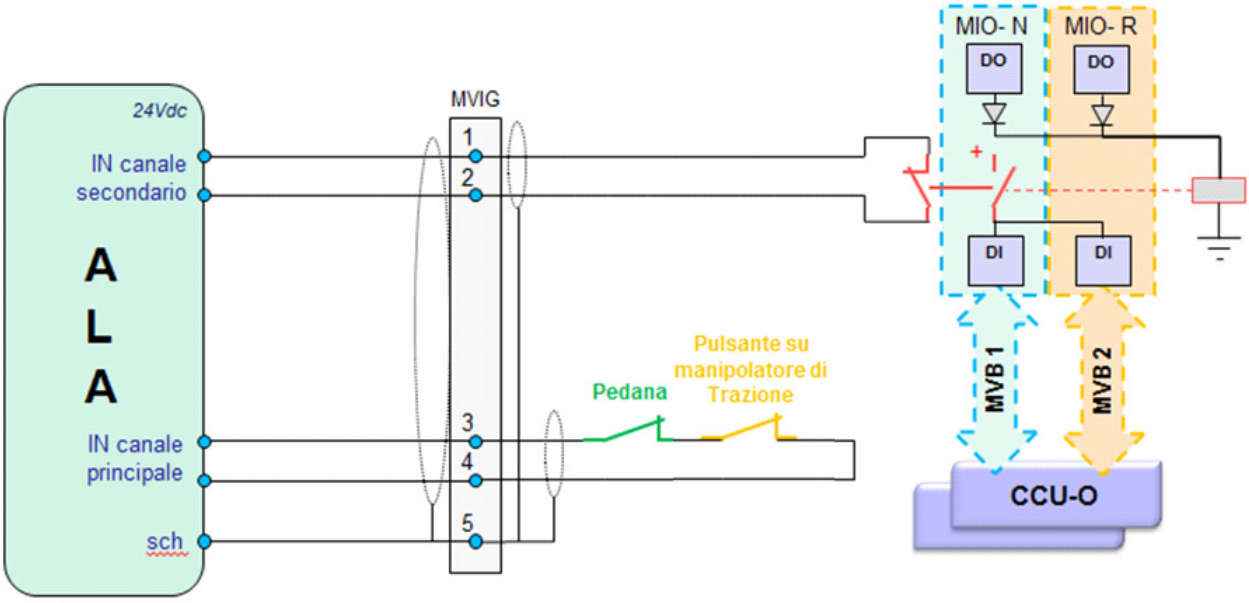


Figure 3: Vigilance device scheme

3.1.1.1.2 Normal condition

The driver confirm its presence and vigilance through the pedal or the pushbutton on the traction lever: there is no buzzer and no brake application
The driver does not confirm its presence or vigilance: the DVD activates the buzzer and there is still no action from the driver it applies the emergency brake

3.1.1.1.3 Failure cases

A. DVD applies the brake when not needed => The train is stopped
B. DVD fails to apply the brake when commanded=>the train is not stopped when needed

3.1.1.1.4 Failure detection

A. Failure self-evident
B. Failure can be detected during the daily test initiated by the driver

3.1.1.1.5 Failure handling

A. The DVD shall be isolated and the system shall be repaired during maintenance
B. If a DVD failure is detected during the daily test the mission is cancelled

3.1.1.1.6 Requirements

ID	2F_09.02.Zefiro-Europe.TRS.105
Description	The driving personnel shall confirm presence and vigilance by activating the safety pedal or pushbuttons.
Safety level	SIL 2

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.02.Zefiro-Europe.CONCEPT.91	"Base" configuraton: the status of the safety pedal (operated/not operated) shall be provided to the DVD (SCMT CU) on the primary chain.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits	HW HMI	SIL 2
2F_09.02.Zefiro-Europe.CONCEPT.189	"Corridors" configuraton: the status of the safety pedal (operated/not operated) shall be provided to the DVD (ATESS CU).	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits	HW HMI	SIL 2
2F_09.02.Zefiro-Europe.CONCEPT.190	"Base" configuraton: the status of the control on the traction brake lever (operated/not operated) shall be provided to the DVD (SCMT CU) on the primary chain.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits	HW HMI	SIL 2
2F_09.02.Zefiro-Europe.CONCEPT.191	"Corridors" configuraton: the status of the control on the traction brake lever (operated/not operated) shall be provided to the DVD(ATESS CU).	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits	HW HMI	SIL 2
2F_09.02.Zefiro-Europe.CONCEPT.624	"Base" configuration: the TCMS shall be able to monitor the vigilance of the driver and to inform the DVD (SCMT CU) through a digital output to the DVD on the "secondary chain".	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		
2F_09.02.Zefiro-Europe.CONCEPT.625	When the TCMS detect driver vigilance (see concept.626) shall enerigize the relevant DO. The signal shall be to grant a LOW signal to the DVD for a time 1 s<t<1.5 s. The signal shall be then deenergized for a time 1 s<t<1.5 s. In case TCMS detect further actions of the driver a new cycle shall be provided by the TCMS to the DVD.	Derived Requirement	4S_09.03.05.-.TCMS Software		
2F_09.02.Zefiro-Europe.CONCEPT.626	When an action, reported in the below list, is detected by TCMS a dedicated DO shall be energized according contept.625 logic. Movement of the traction/brake lever is detected: The sector of 80° shall be divided in 20 sub sector and the lever movement shall be detected when the lever moves from one section to another one. A proper hysteresis shall be defined in order to avoid undue oscillation of the signal. Action on the automatic speed lever (ASC): The ASC moves from position N to position Vel+ or Vel-. Action on the voltage selection selector: The selector is moved from one position to another. Action on LCB lever: The lever moves from neutral position to "OPEN" or "CLOSE". Horn activation: The horn command lever moves from "OFF" position to "HIGH" or to "LOW". Action on IDU display Any driver action on the display such as page change request, or acknowlede of diagnostic events and alarms Action on external light levers	Derived Requirement	4S_09.03.05.-.TCMS Software		

	Movement of levers “Full Beam" (comando luci di profondità) or “Head Lamp" (comando luci di posizione) or “Central Beam" (comando faro centrale) Action on sanding devices push button				
2F_09.02.Zefiro-Europe.CONCEPT.628	A feedback of the TCMS DO “driver vigilance” shall be provided to the TCMS itself for diagnostic purpose.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		
2F_09.02.Zefiro-Europe.CONCEPT.627	A diagnostic event shall be generated by the TCMS in case of incongruence between DO “diver vigilance” and its feedback.	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	

ID	2F_09.02.Zefiro-Europe.TRS.145
Description	Emergency brake shall be applied if the driver does not confirm the presence after first an optical warning and after that a acoustic warning.
Safety level	SIL 2

The MMI for the “Base” configuration is integrated in the ETCS/SCMT DMI and sound generator.

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.02.Zefiro-Europe.CONCEPT.198	“Corridor” configuration : the DVD shall command a dedicated lamp on the driver desk though a digital output when the DVD controls are not operated.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 0
2F_09.02.Zefiro-Europe.CONCEPT.199	“Corridor” configuration : the DVD shall command a dedicated buzzer in the cabin though a digital output when the DVD controls are not operated.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 0
2F_09.02.Zefiro-Europe.CONCEPT.200	“Corridors” configuration: The DVD applies the emergency brake commanding the de energization of the emergency brake loop.	Derived Requirement	4S_09.06.-.Automatic Train Protection Systems		SIL 2

For the base configuration, the emergency brake application is performed through the same interface as for the ATP (DVD is integrated in the ATP): the DVD will command the dedicated pneumatic plate that will cause the brake pipe discharge. Refer to §3.1.2.5

ID	2F_09.02.Zefiro-Europe.TRS.92
Description	The DVD shall be interfaced with the JRU in order to provide data to be recorded
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.02.Zefiro-Europe.CONCEPT.95	The DVD shall provide through MVB to the JRU the DVD EB intervention for recording purpose.	Derived Requirement	4S_09.06.-.Automatic Train Protection Systems		N/A

ID	2F_09.02.Zefiro-Europe.TRS.147
Description	The DVD shall be interfaced with the Cab radio through MVB for the generation of the GSM-R alarm in case of DVD intervention
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.02.Zefiro-Europe.CONCEPT.96	The DVD shall provide through MVB to the cab radio the DVD EB intervention.	Derived Requirement	4S_09.06.-.Automatic Train Protection Systems		N/A
2F_09.02.Zefiro-Europe.CONCEPT.378	The DVD shall provide through digital output to the cab radio the DVD EB intervention.	Derived Requirement	4S_09.06.-.Automatic Train Protection Systems		N/A
2F_09.02.Zefiro-Europe.CONCEPT.	The Cab radio shall send the alarm message when receive the AVD EB intervention	Derived Requirement	4S_10.01.-.Train Radio System		N/A

202					
2F_09.02.Zefiro-Europe.CONCEPT.379	The status of the DVD EB intervention shall be provided to the cab radio.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A

ID	2F_09.02.Zefiro-Europe.TRS.148
Description	The DVD shall get digital input to detect the driver desk status
Safety level	SIL 2

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.02.Zefiro-Europe.CONCEPT.97	"Corridor": The status of the cab (activated/not activated) shall be provided to the DVD.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 2

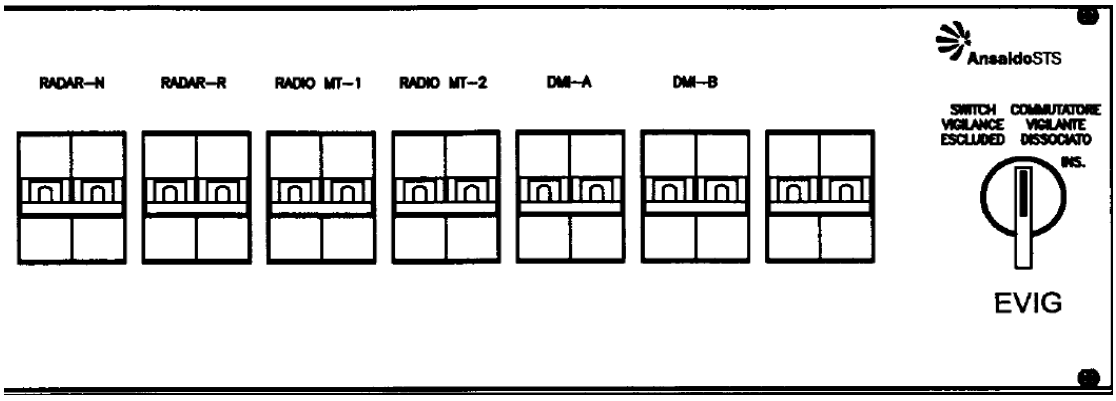
For base configuration DVD is integrated with ATP (same ATP cab status input are used, refer to §3.1.2.4)

3.1.1.2 Driver´s Vigilance Device Isolation

3.1.1.2.1 Function design

Base configuration

The base configuration DVD is integrated into the ATP system. Therefore the isolation of the DVD (in case of failure) equals to isolation of the ATP (ETCS/SCMT).
In the current implementation it is however possible to exclude the vigilance function through a dedicated selector to be foreseen in the area near the driver cab.



When the selector is in the excluded position the vigilance function is excluded and only the Italian specific function "protezione atto partenza" is performed.
When only the function "protezione atto partenza" is active the DVD supervision is limited to the moment when the train start to move, i.e the driver as to acknowledge thorugh the DVD dedicated controls that the train starts to move.
The status of the EVIG is also acquired and recorded by the DIS system.

Corridors configuration

In case of DVD failure the DVD itself applies the emergency brake (fail safe) therefore in this case there shall be the possibility to isolate the device from the EB loop to be able to move the train: the DVD can be isolated through a sealed switch.

3.1.1.2.1.1 **Normal condition**

In case of failure of the DVD the driver isolate it from the brakes.

3.1.1.2.1.2 **Failure cases**

A. DVD is not isolated from the brake when the isolation switch is in isolation position
B. DVD is isolated from the brake when the isolation switch is in normal position

3.1.1.2.1.3 **Failure detection**

A. Failure is self-evident. The train remains braked
B. Failure can be detected during the daily test initiated by the driver. This failure however can be expected only after switch movement from isolation to normal position, therefore after the action the execution of the test is expected.

3.1.1.2.1.4 **Failure handling**

A. To release brake application pneumatic valve could be bypassed
B. The failure shall be repaired during maintenance

3.1.1.2.2 Requirements

ID	2F_09.02.Zefiro-Europe.TRS.111
Description	It shall be possible to Bypass the EB applied by the DVD in case of a faulty DVD system.
Safety level	SIL 2

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.02.Zefiro-Europe.CONCEPT.310	"Corridor" configuration: a sealed switch on the cab shall be foreseen to bypass the DVD brake application in the emergency brake loop.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits	HW HMI	SIL 2
2F_09.02.Zefiro-Europe.CONCEPT.311	"Corridor" configuration: the TCMS shall acquire the status of the DVD isolation Switch though a digital input.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits	HW HMI	SIL 0
2F_09.02.Zefiro-Europe.CONCEPT.312	"Corridor" configuration: the TCMS shall warn the driver throgugh the IDU when the DVD is isolated	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_09.02.Zefiro-Europe.CONCEPT.313	"Corridor" configuration: the TCMS shall communicate to the train recorder when the DVD is isolated	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.02.Zefiro-Europe.CONCEPT.314	"Corridor" configuration: the DVD shall acquire the status of the DVD isolation Switch though a digital input (to inhibit its functions)	Derived Requirement	4S_09.06.-.Automatic Train Protection Systems		SIL 2
2F_09.02.Zefiro-Europe.CONCEPT.381	A switch named EVIG on the cab shall be foreseen to exclude the vigilance function	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits	HW HMI	SIL 2
2F_09.02.Zefiro-Europe.CONCEPT.382	The status of the switch EVIG shall be provided to the DVD (SCMT CU).	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 2
2F_09.02.Zefiro-Europe.CONCEPT.383	The status of the switch EVIG shall be provided to the DIS for recording purpose	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 0

ID	2F_09.02.Zefiro-Europe.TRS.254
Description	The DVD test shall be started manually by the driver in the active cab.
Safety level	SIL 0

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.02.Zefiro-Europe.CONCEPT.315	"Corridor" configuration: The diagnostic display shall foresee a soft key to command the Dvd test.	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0

2F_09.02.Zefiro-Europe.CONCEPT.316	"Corridor" configuration: The TCMS shall command a Dvd test though MVB when the driver pushes the relevant soft key on the IDU.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

3.1.2 Automatic Train Protection System (ATP)

3.1.2.1 Visualization

3.1.2.1.1 Function design

3.1.2.1.1.1 **Normal condition**

The ETCS DMI shall visualize the time received by the DIS system (GPS time).
The speed is normally shown on the ETCS DMI.
In case of ATP system failure and consequent isolation the speed will be displayed on the IDU.

3.1.2.1.1.2 **Failure cases**

A. ATP system wrongly detected as isolated by the TCMS => speed is displayed contemporaneously on ATP DMI and on IDU
B. ATP system wrongly detected as not isolated by the TCMS => speed signal is not available to the driver

3.1.2.1.1.3 **Failure detection**

A. Failure self-evident
B. Failure self.evident

3.1.2.1.1.4 **Failure handling**

A. Failure isolated and repaired during maintenance
B. Failure isolated and repaired during maintenance

3.1.2.1.2 Requirements

ID	2F_09.02.Zefiro-Europe.TRS.154
Description	The train speed shall be displayed on the diagnostic display in case of absence or degradation of the ATP system.
Safety level	SIL 2

Nr	**Description**	**Type**	**Derived to**	**User Interface**	**SIL**
2F_09.02.Zefiro-Europe.CONCEPT.317	If the ATP is isolated by the isolation switch (CEA) the train speed shall be displayed on the IDU.	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0
2F_09.02.Zefiro-Europe.CONCEPT.384	The TCMS shall acquire the status of the ATP isolation switch (CEA). Refer to RD 1 sheet 52.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 0
2F_09.02.Zefiro-Europe.CONCEPT.385	The CEA status shall be recorded by the DIS/JRU	Derived Requirement	4S_09.06.-.Automatic Train Protection Systems		N/A
2F_09.02.Zefiro-Europe.CONCEPT.386	The status of the CEA shall be provided to the DIS/JRU	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A

ID	2F_09.02.Zefiro-Europe.TRS.174
Description	The ATP system shall provide the time to the train for syncronization to all equipment onboard.
Safety level	SIL 0

The time for on-board synchronization will come from DIS and be sent to TCMS. TCMS will use the MCG via SNTP to do the synchronization.

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.02.Zefiro-Europe.CONCEPT.617	The DIS shall send the time in UTF format to TCMS for time synchronization on-board.	Derived Requirement	4S_09.06.-.Automatic Train Protection Systems		SIL 0
2F_09.02.Zefiro-Europe.CONCEPT.618	The TCMS shall use the time from the DIS for on-board time synchronization	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

3.1.2.2 ATP Direction control

3.1.2.2.1 Function design

The ATP monitor the direction selected by the driver and the actual train movement: in case of incoherence it applies the emergency brake.

3.1.2.2.1.1 **Normal condition**

The selected direction is neutral (no selection), the train moves forward or backward => ATP emergency brake application
The selected direction is forward, the train moves backward => ATP emergency brake application
The selected direction is backward, the train moves forward => ATP emergency brake application

3.1.2.2.1.2 **Failure cases**

A. Loss of ATP emergency brake application

3.1.2.2.1.3 **Failure detection**

A. Failure detected during system power on self-test

3.1.2.2.1.4 **Failure handling**

A. Failure isolated and repaired during maintenance

3.1.2.2.2 Requirements

ID	2F_09.02.Zefiro-Europe.TRS.158
Description	The requirements of signals necessary for ATP to control the driving direction of the train during ATP system supervision shall be defined. The defined driving directions could be: - Forward Direction - Backward Direction
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.02.Zefiro-Europe.CONCEPT.318	“Base” configuration: the TCMS shall provide to the ETCS the signal position Forward of the direction controller through MVB	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.02.Zefiro-Europe.CONCEPT.319	“Base” configuration: the TCMS shall provide to the ETCS the signal position Backward of the direction controller through MVB	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.02.Zefiro-Europe.CONCEPT.320	“corridors” configuration:CTC shall provide to the ETCS the signal position forward of the direction controller through digital input	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 2
2F_09.02.Zefiro-Europe.CONCEPT.	“corridors” configuration: CTC shall provide to the ETCS the signal position backward of the direction	Derived Requirement	4S_09.02.09.-.Conventional Train		SIL 2

321	controller through digital input		Control - Circuits		
2F_09.02.Zefiro-Europe.CONCEPT.110	”Base” configuration: If the direction controller is set to Neutral then Forward and Backward direction signals over the MVB shall be reset to the ATP.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.02.Zefiro-Europe.CONCEPT.377	“corridors” configuration: If the direction controller is set to Neutral then Forward and Backward direction signals shall be reset to the ATP.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 2

3.1.2.3 ATP Isolation

3.1.2.3.1 Function design

In case of ATP failure the ATP itself applies the emergency brake (fail safe) therefore in this case there shall be the possibility to isolate the device to be able to move the train: the ATP can be isolated through a sealed switch.

3.1.2.3.1.1 **Normal condition**

In case of failure of the ATP the driver isolate it from the brakes.

3.1.2.3.1.2 **Failure cases**

A. ATP is not isolated from the brake when the isolation switch is in isolation position
B. ATP is isolated from the brake when the isolation switch is in normal position

3.1.2.3.1.3 **Failure detection**

A. Failure is self-evident. The train remains braked
B. Failure can be detected during power on test. This failure however can be expected only after switch movement from isolation to normal position, therefore after the action the execution of the test is expected.

3.1.2.3.1.4 **Failure handling**

A. To release brake application pneumatic valve could be bypassed
B. The failure shall be repaired during maintenance

3.1.2.3.2 Requirements

ID	2F_09.02.Zefiro-Europe.TRS.120
Description	ATP isolation shall be possible using a sealed isolation switch.
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.02.Zefiro-Europe.CONCEPT.322	“Corridor” configuration: a sealed switch on the cab shall be foreseen to bypass the ATP brake application in the emergency brake loop	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits	HW HMI	SIL 3
2F_09.02.Zefiro-Europe.CONCEPT.323	“Corridor” configuration: the ATP shall acquire the status of the ATP isolation Switch though a digital input	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits	HW HMI	SIL 3
2F_09.02.Zefiro-Europe.CONCEPT.324	“Corridor” configuration: the TCMS shall warn the driver through the IDU when the ATP is isolated	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	SIL 0
2F_09.02.Zefiro-Europe.CONCEPT.111	“Corridor” configuration: the TCMS shall communicate to the train recorder when the ATP is isolated	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

3.1.2.4 ATP Start up Mission during Cab activation

3.1.2.4.1 Function design

3.1.2.4.1.1 **Normal condition**

The system shall propose to the driver the start of mission when the cab is activated

3.1.2.4.1.2 **Failure cases**

A. ATP does not detect the cab activation=> the mission shall be cancelled

3.1.2.4.1.3 **Failure detection**

A. Failure is self-evident

3.1.2.4.1.4 **Failure handling**

A. Failure isolated and repaired during maintenance

3.1.2.4.2 Requirements

ID	2F_09.02.Zefiro-Europe.TRS.124
Description	ATP shall automatically start if cab has been activated. If cab has been activated ATP in the activated cab shall get a signal which allows the ATP to start its Mission.
Safety level	SIL 0

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.02.Zefiro-Europe.CONCEPT.325	Base configuration: the status " cab Active shall be provided to the ATP through 1 switch. Refer to RD1 sheet 52 and RD3 §2.1.6.3	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_09.02.Zefiro-Europe.CONCEPT.115	Base configuration: the status "cab not Active" shall be provided to the ATP through 1 switch. Refer to RD1 sheet 52 and RD3 §2.1.6.3	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A

3.1.2.5 ATP Emergency Brake control

3.1.2.5.1 Function design

3.1.2.5.1.1 **Normal condition**

Base configuration:
The ATP applies the emergency brake commanding directly the dedicated local electro valves: the brake pipe is then discharged from the active cab

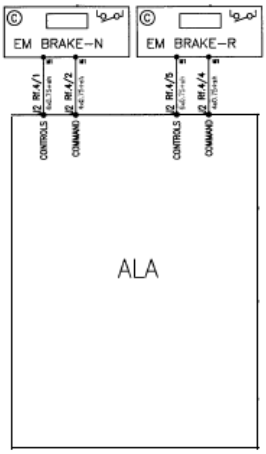


Figure 3: ALA- Pneumatic plates interface

In parallel to the pneumatic plate command the ATP commands also the inhibition of the refilling of the brake pipe: this command will be used to de energize the emergency brake loop.

Corridors configuration:
The ATP applies the emergency brake commanding the de energization of the emergency brake loop.

3.1.2.5.1.2 Failure cases

A. Loss to de energize the emergency brake loop => the brake pipe will be discharged only from the active cab therefore the equivalent time of the brake application will be greater

3.1.2.5.1.3 Failure detection

A. Failure detected by the TCMS

3.1.2.5.1.4 Failure handling

A. Failure isolated and repaired during maintenance

3.1.2.5.2 Requirements

ID	2F_09.02.Zefiro-Europe.TRS.130
Description	ATP shall be integrated in the train's electrical and pneumatical circuits in such a manner that it can order Emergency Brake.
Safety level	SIL 4

ID	2F_09.02.Zefiro-Europe.TRS.170
Description	When ordered by ATP Emergency brake shall be applied in the whole train by cutting the power supply from the magnet valve (reverse) of the pneumatic brake.
Safety level	SIL 4

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.02.Zefiro-	Base configuration: the ATP will order emergency	Derived	4S_09.06.-.Automatic		SIL 4

Europe.CONCEPT.326	brake commanding its dedicated pneumatic plate	Requirement	Train Protection Systems		
2F_09.02.Zefiro-Europe.CONCEPT.327	Base configuration: in parallel to the command to the Pneumatic plate the ATP will command the Brake pipe refilling inhibition through a digital output. Refer to RD1 sheet 52 and RD3 §2.1.9.3	Derived Requirement	4S_09.06.-.Automatic Train Protection Systems		SIL 4
2F_09.02.Zefiro-Europe.CONCEPT.328	Base configuration: when the ATP command the brake pipe refilling inhibition Emergency brake shall be applied in the whole train by de-energising the Emergency brake loop .	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 4
2F_09.02.Zefiro-Europe.CONCEPT.117	Corridors configuration: The ATP applies the emergency brake commanding the de energization of the emergency brake loop.	Derived Requirement	4S_09.06.-.Automatic Train Protection Systems		SIL 4
2F_09.02.Zefiro-Europe.CONCEPT.329	If ATP orders Emergency brake an event shall be set.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

ID	2F_09.02.Zefiro-Europe.TRS.182
Description	In parallell with the pneumatic Emergency brake order to the Braking system an electrical signal shall be sent to the Braking system to apply Emergency brake.
Safety level	SIL 3

The requirement 2F_09.02.Zefiro-Europe.TRS.182 will not be considered for this release of the SFRS. The emergency brake interface between ATP and Braking system will be further discussed and developed in next release of this document.

3.1.2.6 ATP Service Brake control

3.1.2.6.1 Function design

3.1.2.6.1.1 **Normal condition**

The ATP will command the service brake when needed (First Line of Intervention), the TCMS will forward the request to the brake system that shall apply the service brake.
Note: in case the service brake command will not be applied or in case the speed reduction is not enough to avoid crossing the second line of intervention the ATP will command the emergency brake.

3.1.2.6.1.2 **Failure cases**

A. Service brake not applied when commanded by the ATP

3.1.2.6.1.3 **Failure detection**

A. Failure self-evident: EB applied by the ATP

3.1.2.6.1.4 **Failure handling**

A. Failure isolated and repaired during maintenance

3.1.2.6.2 Requirements

ID	2F_09.02.Zefiro-Europe.TRS.132
Description	ATP shall be able to order Service Brake.
Safety level	SIL 0

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.02.Zefiro-Europe.CONCEPT.	Corridors configuration: The ATP shall command the full service brake through MVB.	Derived Requirement	4S_09.06.-.Automatic Train Protection		SIL 0

330			Systems		
2F_09.02.Zefiro-Europe.CONCEPT.331	Base configuration: The ATP shall command the full service brake through MVB.	Derived Requirement	4S_09.06.-.Automatic Train Protection Systems		SIL 0
2F_09.02.Zefiro-Europe.CONCEPT.119	The TCMS shall apply service brake when ordered by ATP according to Brake blending rules defined in the Brake SFRS	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.02.Zefiro-Europe.CONCEPT.387	The Italian national ATP (SCMT) shall command the service brake through a digital output (free voltage contact)	Derived Requirement	4S_09.06.-.Automatic Train Protection Systems		SIL 0
2F_09.02.Zefiro-Europe.CONCEPT.388	The TCMS shall acquire the service brake command	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 0
2F_09.02.Zefiro-Europe.CONCEPT.389	The TCMS shall apply service brake when ordered by ATP (SCMT)	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

3.1.2.7 Traction Cut-off

3.1.2.7.1 Function design

3.1.2.7.1.1 **Normal condition**

The ATP will command the SW traction block when needed (phase and voltage transition, speed limitation), the TCMS will forward the request to the traction system that shall remove the traction effort.
Also the DIS system shall be able to block the traction in specific circumstances (for example in case the speed is over the admitted threshold and the ATP is isolated, i.e the switch CEA has been operated)

3.1.2.7.1.2 **Failure cases**

A. Traction effort wrongly not removed => the MCB will be opened with traction effort

3.1.2.7.1.3 **Failure detection**

A. Failure not detected

3.1.2.7.1.4 **Failure handling**

A. Failure isolated and repaired during maintenance

3.1.2.7.2 Requirements

ID	2F_09.02.Zefiro-Europe.TRS.136
Description	If ATP system orders brake (Emergency Brake / Service Brake) ATP will send a signal which shall lead to software block of traction demand in the whole train.
Safety level	SIL 0

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.02.Zefiro-Europe.CONCEPT.332	Base configuration: The ATP shall be able to command SW traction block when necessary through a digital output. Refer to RD1 sheet 52 and RD3 §2.1.9.1	Derived Requirement	4S_09.06.-.Automatic Train Protection Systems		SIL 0
2F_09.02.Zefiro-Europe.CONCEPT.392	Base configuration: TCMS shall acquire the SW traction block command	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 0
2F_09.02.Zefiro-Europe.CONCEPT.333	Corridors configuration: the ATP shall be able to command SW traction block when necessary (phase transition) through MVB.	Derived Requirement	4S_09.06.-.Automatic Train Protection Systems		SIL 0

2F_09.02.Zefiro-Europe.CONCEPT.121	If ATP SW traction block command is applied AND the switch CEA is not operated then Traction block shall be applied in the whole train.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.02.Zefiro-Europe.CONCEPT.393	The DIS shall be able to command SW traction block (trough a digital output)	Derived Requirement	4S_09.06.-.Automatic Train Protection Systems		SIL 0
2F_09.02.Zefiro-Europe.CONCEPT.394	The TCMS shall acquire the SW traction block command from the DIS	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 0
2F_09.02.Zefiro-Europe.CONCEPT.398	A sealed switch named CETD on the cab shall be foreseen to bypass the DIS SW traction block	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits	HW HMI	
2F_09.02.Zefiro-Europe.CONCEPT.397	The DIS shall acquire the status of the switch CETD for recording purpose	Derived Requirement	4S_09.06.-.Automatic Train Protection Systems		
2F_09.02.Zefiro-Europe.CONCEPT.396	The TCMS shall acquire the status of the switch CETD	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		
2F_09.02.Zefiro-Europe.CONCEPT.395	When the switch CETD is operated the lamp “avaria DIS” shall be switched on (refer to 2F_09.02.Zefiro-Europe.CONCEPT.334)	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		
2F_09.02.Zefiro-Europe.CONCEPT.399	If DIS SW traction block command is applied by the DIS related to the active cab AND the switch CETD is not operated then Traction block shall be applied in the whole train by the TCMS. Note: traction block command of DIS of not active cabs shall ne ignored by the TCMS.	Derived Requirement	4S_09.03.05.-.TCMS Software		

3.1.2.8 Main Circuit breaker command

3.1.2.8.1 Function design

3.1.2.8.1.1 **Normal condition**

The ATP will command the main circuit breaker command when needed (phase and voltage transition, speed limitation); the TCMS will forward the request to the traction system that will command the MCB opening.
The ATP will command the MCB opening through two different command for redundancy (MVB and Digital output) to guarantee the command execution. It is then a “Train” task to assure the execution of the command: the digital output can be acquired by the TCMS and the command managed by the TCMS SW or it can be directly used to drive a Trail line to command the MCB opening via HW.

3.1.2.8.1.2 **Failure cases**

3.1.2.8.1.3 **Failure detection**

3.1.2.8.1.4 **Failure handling**

3.1.2.8.2 Requirements

ID	2F_09.02.Zefiro-Europe.TRS.256
Description	The ATP shall give additional information to open the main circuit breaker. On leaving the separation section the ATP shall cause main circuit breaker to be closed.
Safety level	SIL 0

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.02.Zefiro-	Base configuration: The ATP shall be able to	Derived	4S_09.06.-.Automatic		SIL 0

Europe.CONCEPT.433	command main circuit breakers opening when necessary thorugh a digital output. Refer to RD1 sheet 52 and RD3 §2.1.9.7	Requirement	Train Protection Systems		
2F_09.02.Zefiro-Europe.CONCEPT.437	The TCMS shall acquire the MCB opening command	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 2
2F_09.02.Zefiro-Europe.CONCEPT.436	The TCMS shall command the MCB opening when ordered by ATP (SCMT) via digital output	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.02.Zefiro-Europe.CONCEPT.435	In redundancy to the digital command the ATP shall be able to command main circuit breakers opening thorugh the MVB	Derived Requirement	4S_09.06.-.Automatic Train Protection Systems		SIL 0
2F_09.02.Zefiro-Europe.CONCEPT.434	The TCMS shall command the MCB opening when ordered by ATP (SCMT) via MVB	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

3.1.2.9 Phase Transition

3.1.2.9.1 Function design

3.1.2.9.1.1 **Normal condition**

The phase transition in FS is completely automatic while in SR and when the ATP is isolated it shall be realized manually.
The ATP will command to the TCMS the SW traction block and MCB opening in advance to the neutral section and will remove these commands at the end of the neutral section(in multiple operation: commands are removed when both units have left the neutral section).
In order to reduce the time with the SW traction block and open MCB the TCMS shall also communicate to the ATP which is the raised 25kV pantograph (front or rear pantograph): the ATP will then command the SW traction block and MCB opening before the active pantograph reaches the neutral section and will give the command MCB closure and traction reestablishment when the active pantograph has left the neutral section.
In case of multiple unit the ATP commands to the second unit will be shifted by the TCMS when the travelled distance matches the distance between the active pantograph of the first unit and the active pantograph of the second unit.

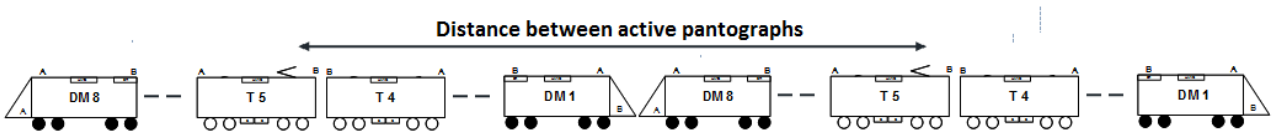


Figure 4: distance from active pantographs in multiple units

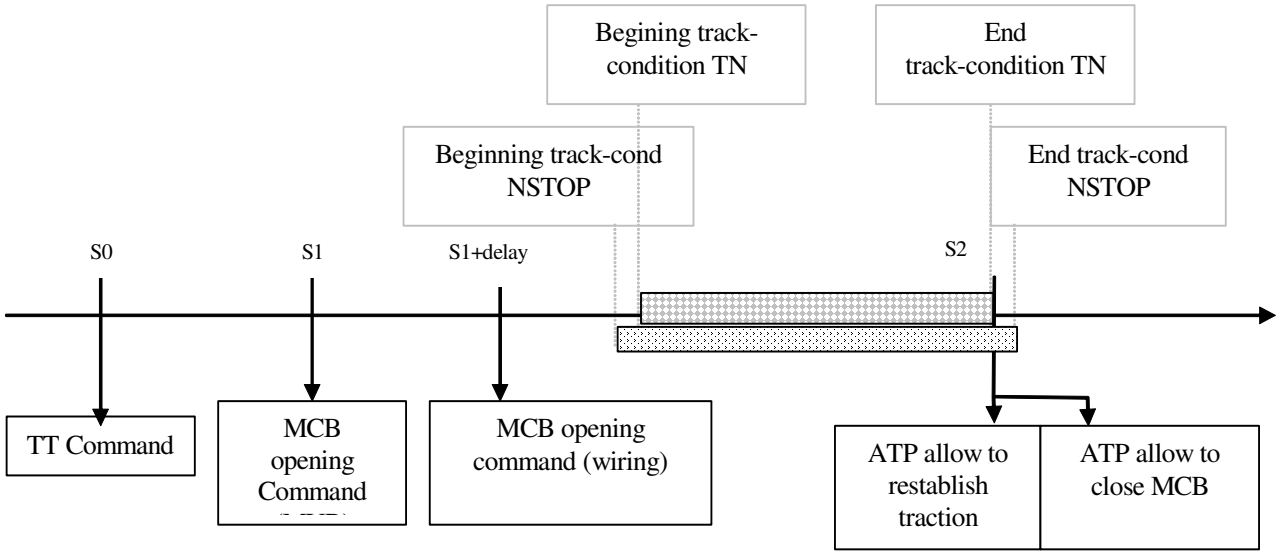


Figure 5: Phase transition sequence

3.1.2.9.1.2 **Failure cases**

3.1.2.9.1.3 **Failure detection**

3.1.2.9.1.4 **Failure handling**

3.1.2.9.2 Requirements

ID	2F_09.02.Zefiro-Europe.TRS.258
Description	The phase separation sections and separation section sequences shall be activated by External sensors e.g. ATP or track side magnets.
Safety level	SIL 0

The ATP will compute the location where to cut off the traction and will command it as described in §3.1.2.7
The ATP will compute the location where to open the MCB and will command it as described in §3.1.2.8
The computation of the location to command and revoke the SW traction block and MCB opening command is based on the fact that the ETCS know which is the active pantograph (front or back pantograph)

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.02.Zefiro-Europe.CONCEPT.451	The TCMS shall inform the ATP through MVB which is the active pantograph	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

When the condition for the traction restoration are met the ATP will remove the SW traction block and MCB opening command

3.1.2.10 Voltage Transition

3.1.2.10.1 Function design

3.1.2.10.1.1 **Normal condition**

The voltage transition in FS is automatic and it requires to the driver only to select the new voltage throught the line voltage selector on the driver desk at the end of the neutral section.
In SR and when the ATP is isolated it shall be realized manually.
The ATP will command to the TCMS the SW traction block, MCB opening and pantograph lowering in advance to the neutral section and will remove these commands at the end of the neutral section(in multiple operation: commands are removed when both units have left the neutral section.).

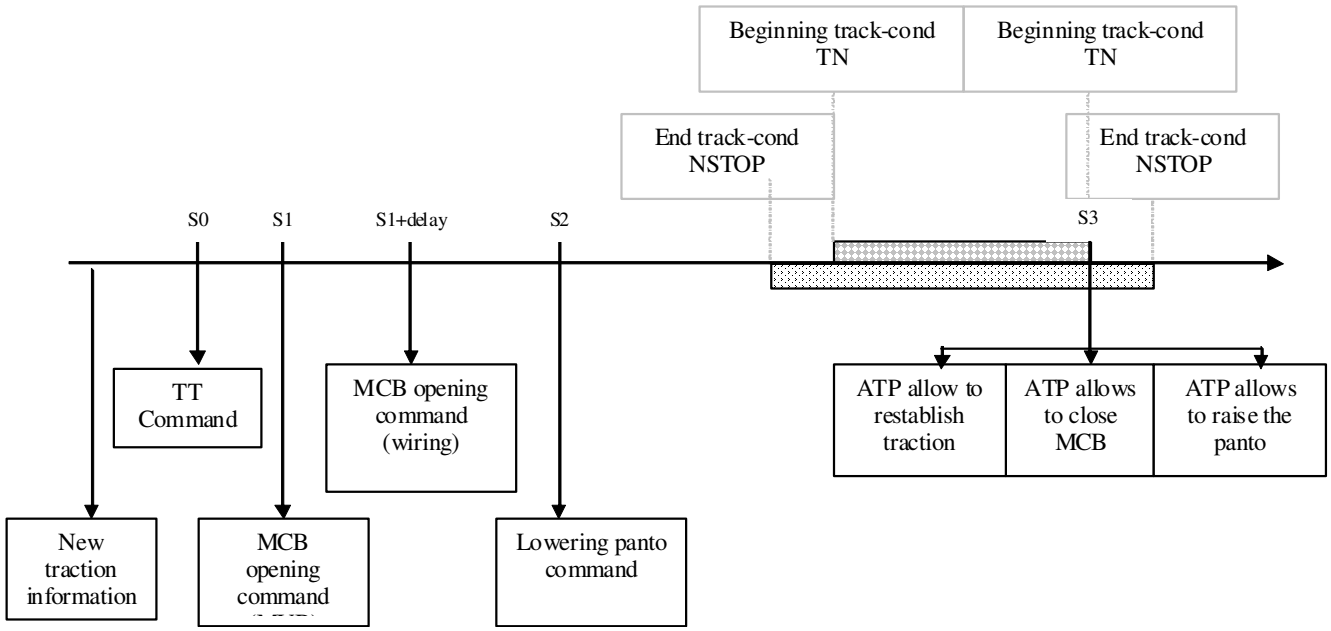


Figure 6: Voltage transition sequence

3.1.2.10.1.2 **Failure cases**

3.1.2.10.1.3 **Failure detection**

3.1.2.10.1.4 **Failure handling**

3.1.2.10.2 Requirements

ID	2F_09.02.Zefiro-Europe.TRS.260
Description	Where phase separation sections require pantographs to be lowered and subsequently raised, these additional actions are permitted to be automatically intiated by ATP. These functions shall respond to input signals from the control-command and signalling system.
Safety level	

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.02.Zefiro-	The ATP shall be able to send the new voltage	Derived	4S_09.06.-.Automatic		SIL 0

Europe.CONCEPT.457	value to the TCMS via MVB (change power command)	Requirement	Train Protection Systems		

The ATP will compute the location where to cut off the traction
The ATP will compute the location where to open the MCB and will command it as described in §3.1.2.8

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.02.Zefiro-Europe.CONCEPT.576	The ATP will compute the location where to lower the pantograph and will command it through the MVB (Raising pantograph command)	Derived Requirement	4S_09.06.-.Automatic Train Protection Systems		SIL 0
2F_09.02.Zefiro-Europe.CONCEPT.460	The TCMS shall lower the pantograph when commanded by the ATP	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

When the condition for the traction restoration are met the ATP will remove the SW traction block, MCB opening and lower the pantograph command

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.02.Zefiro-Europe.CONCEPT.463	If the ATP removes the SW traction block, MCB opening and lower the pantograph command the TCMS shall raise the pantograph if the driver has correctly selected the voltage type and if all safety condition of its raising are met.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.02.Zefiro-Europe.CONCEPT.464	If the ATP removes the SW traction block, MCB opening and lower the pantograph command and the pantograph has been raised the TCMS shall close the MCB and restore the traction only if all safety condition are met.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_09.02.Zefiro-Europe.CONCEPT.465	If the line voltage system selected by the driver does not correspond to the line voltage system reported by the ATP an event shall be set and the pantograph shall not be allowed to be raised.	Derived Requirement	4S_09.03.05.-.TCMS Software		SIL 0

3.1.3 Juridical Recorder Unit (JRU)

3.1.3.1 Function design

3.1.3.1.1 **Normal condition**

The train recorder shall record all signal coming from the train through MVB (TCMS) or through digital input

3.1.3.1.2 **Failure cases**

N/A

3.1.3.1.3 **Failure detection**

N/A

3.1.3.1.4 **Failure handling**

N/A

3.1.3.2 Requirements

ID	2F_09.02.Zefiro-Europe.TRS.169
Description	Signals to be recorded as a minimum: · "Clear to proceed" from the ATP system · "Restrictive signal aspect" from the ATP system · "Acknowledgement" from the ATP system · Time · Speed

	· Brake applied · EB applied · ATP cab switch · The interaction between the driver and the ATP system in terms of actions done by the personnel and of communications sent to the driver by means of visual and/or acoustic signals · The data which the onboard technological system exchanges with the ground sub-systems with which it communicates · Distance · The data that the ATP exchanges with the equipment and systems of the train · The data pertaining to the ATP functional status.
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.02.Zefiro-Europe.CONCEPT.126	The TCMS shall provide to the DIS data to be recorded though MVB according to RD 8 and RD 9.	Derived Requirement	4S_09.03.05.-.TCMS Software		N/A
2F_09.02.Zefiro-Europe.CONCEPT.619	The DIS shall record data received through MVB according to RD 8	Derived Requirement	4S_09.06.-.Automatic Train Protection Systems		
2F_09.02.Zefiro-Europe.CONCEPT.620	The status of digital input shall be provided to the DIS according to RD 8.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		
2F_09.02.Zefiro-Europe.CONCEPT.621	The DIS shall record digital input status according to RD 8.	Derived Requirement	4S_09.06.-.Automatic Train Protection Systems		

ID	2F_09.02.Zefiro-Europe.TRS.175
Description	A faulty JRU shall be reported to the driver.
Safety level	N/A

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.02.Zefiro-Europe.CONCEPT.334	The DIS shall command a lamp “avaria DIS” on the driver desk in case of failure	Derived Requirement	4S_09.06.-.Automatic Train Protection Systems		N/A
2F_09.02.Zefiro-Europe.CONCEPT.622	The lamp “avaria DIS” shall be switched on also in case of DIS circuit breaker is tripped	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		
2F_09.02.Zefiro-Europe.CONCEPT.129	An event shall be generated if the JRU is faulty	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_09.02.Zefiro-Europe.CONCEPT.335	The DIS shall command a lamp on the driver desk in case of recorder memory is full (70% of memory)	Derived Requirement	4S_09.06.-.Automatic Train Protection Systems		N/A
2F_09.02.Zefiro-Europe.CONCEPT.336	An event shall be generated in case of recorder memory is full	Derived Requirement	4S_09.03.05.-.TCMS Software	EVENT	N/A
2F_09.02.Zefiro-Europe.CONCEPT.467	The DIS shall be switched on as soon as the battery are connected.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_09.02.Zefiro-Europe.CONCEPT.623	The status of the bus 180 m shall be provided to the DIS/JRU for the management of the sleeping function.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		
2F_09.02.Zefiro-Europe.CONCEPT.469	The DIS/JRU shall acquire the status of the cab (active/not active)	Derived Requirement	4S_09.06.-.Automatic Train Protection Systems		N/A
2F_09.02.Zefiro-Europe.CONCEPT.468	The status of the cab (activated/not activated) shall be provided to the DIS/JRU	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A

3.1.4 Cab Radio

3.1.4.1 Function design

3.1.4.1.1 **Normal condition**

The cab is equipped with a GSM-R radio for train ground voice communication.The radio will be connected to the intercom and PA system and will have proper interface to the vigilance device for the alarm generation in case of DVD intervention.
The radio shall be powered as soon as the battery contactors are closed but it shall be possible to power it also when the train is off through a dedicated switch.
The failure of the radio will be clearly indicated on the driver desk.

3.1.4.1.2 Failure Cases

N/A

3.1.4.1.3 Failure Detection

N/A

3.1.4.1.4 Failure Handling

N/A

3.1.4.2 Requirements

ID	2F_09.02.Zefiro-Europe.TRS.262
Description	The cab radio shall be compliant to RFI standard RFI DTC CSI SR OR 10 003 A, “Sistema Tecnologico di Bordo - Apparato "Cab Radio": requisiti nazionali ad integrazione dei requisiti europei”
Safety level	N/A

Power management

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.02.Zefiro-Europe.CONCEPT.483	The cab radio shall be switched on as soon as the battery are connected.	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_09.02.Zefiro-Europe.CONCEPT.484	There shall be a switch in the cabin in order to power on the cab radio when the battery are disconnected	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A

Alarm lamp

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.02.Zefiro-Europe.CONCEPT.486	In case the cab radio is not powered or in case of cab radio major fault the cab radio shall provide this information trough a contact free of voltage (digital output Cab radio not available, alarm light mng, refer to RD 7 sheet 6)	Derived Requirement	4S_10.01.-.Train Radio System		SIL 0
2F_09.02.Zefiro-Europe.CONCEPT.487	When the cab radio is not available a light indication shall be switched on the driver desk	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A
2F_09.02.Zefiro-Europe.CONCEPT.488	The alarm lamp status shall be provided to the DIS/JRU	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		N/A

Alarm message

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.02.Zefiro-Europe.CONCEPT.493	The cab radio shall acquire the EB status sent from the DVD (refer to §3.1.1 and to to RD 7 sheet 6, input DSD alarm input).	Derived Requirement	4S_10.01.-.Train Radio System		N/A

Cab status

Nr	Description	Type	Derived to	User Interface	SIL
2F_09.02.Zefiro-Europe.CONCEPT.490	The MMI of the cab radio shall acquire the status of the cab	Derived Requirement	4S_10.01.-.Train Radio System		SIL 0
2F_09.02.Zefiro-Europe.CONCEPT.491	The status of the cab (activated/not activated) shall be provided to the MMI of the radio	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 0

4 Signal Interface

4.1 I/O Interface (base configuration)

4.1.1 Supplier system

Input (ATP/DVD):

Nr	Functional name	Type	Linked to requirement
2F_09.02.Zefiro-Europe.CONCEPT.28	SafetyPedal	L	2F_09.02.Zefiro-Europe.CONCEPT.91
2F_09.02.Zefiro-Europe.CONCEPT.344	ManipulatorControl	L	2F_09.02.Zefiro-Europe.CONCEPT.190
2F_09.02.Zefiro-Europe.CONCEPT.538	EVIG	L	2F_09.02.Zefiro-Europe.CONCEPT.382
2F_09.02.Zefiro-Europe.CONCEPT.339	CabActive	L	2F_09.02.Zefiro-Europe.CONCEPT.325
2F_09.02.Zefiro-Europe.CONCEPT.338	CabNotActive	L	2F_09.02.Zefiro-Europe.CONCEPT.115

Inputs (DIS):

Nr	Functional name	Type	Linked to requirement
2F_09.02.Zefiro-Europe.CONCEPT.337	EVIG	L	2F_09.02.Zefiro-Europe.CONCEPT.383
2F_09.02.Zefiro-Europe.CONCEPT.347	CETD	L	2F_09.02.Zefiro-Europe.CONCEPT.397
2F_09.02.Zefiro-Europe.CONCEPT.539	CEA	L	2F_09.02.Zefiro-Europe.CONCEPT.386
2F_09.02.Zefiro-Europe.CONCEPT.540	Cab Status	L	2F_09.02.Zefiro-Europe.CONCEPT.468
2F_09.02.Zefiro-Europe.CONCEPT.541	Pneumatic plate 1 status	L	
2F_09.02.Zefiro-Europe.CONCEPT.542	Pneumatic plate 2 status	L	

Input (cab radio):

Nr	Functional name	Type	Linked to requirement
2F_09.02.Zefiro-Europe.CONCEPT.546	DVD intervention	L	2F_09.02.Zefiro-Europe.CONCEPT.379
2F_09.02.Zefiro-Europe.CONCEPT.547	Cab status	L	2F_09.02.Zefiro-Europe.CONCEPT.491

Outputs (ATP):

Nr	Functional name	Type	Linked to requirement
2F_09.02.Zefiro-Europe.CONCEPT.553	DVD intervention	L	2F_09.02.Zefiro-Europe.CONCEPT.378
2F_09.02.Zefiro-Europe.CONCEPT.559	Brake pipe refilling inhibition	L	2F_09.02.Zefiro-Europe.CONCEPT.327
2F_09.02.Zefiro-Europe.CONCEPT.558	Service brake command	L	2F_09.02.Zefiro-Europe.CONCEPT.387
2F_09.02.Zefiro-Europe.CONCEPT.557	SW traction block	L	2F_09.02.Zefiro-Europe.CONCEPT.332
2F_09.02.Zefiro-Europe.CONCEPT.556	Main Circuit breaker command	L	2F_09.02.Zefiro-Europe.CONCEPT.433
2F_09.02.Zefiro-Europe.CONCEPT.555	Cab radio not available	L	

Output (DIS):

Nr	Functional name	Type	Linked to requirement
2F_09.02.Zefiro-Europe.CONCEPT.561	SW traction block DIS	L	2F_09.02.Zefiro-Europe.CONCEPT.393
2F_09.02.Zefiro-Europe.CONCEPT.563	Avaria DIS	L	2F_09.02.Zefiro-Europe.CONCEPT.334
2F_09.02.Zefiro-Europe.CONCEPT.564	70%memory full	L	2F_09.02.Zefiro-Europe.CONCEPT.335

Outputs (Cab Radio):

Nr	Functional name	Type	Linked to requirement
2F_09.02.Zefiro-Europe.CONCEPT.568	Cab radio major fault	L	2F_09.02.Zefiro-Europe.CONCEPT.486

4.1.2 TCMS SW

Inputs:

Nr	Functional name	Type	Linked to requirement
2F_09.02.Zefiro-Europe.CONCEPT.30	CEA Status	L	2F_09.02.Zefiro-Europe.CONCEPT.384
2F_09.02.Zefiro-Europe.CONCEPT.569	Service brake	L	2F_09.02.Zefiro-Europe.CONCEPT.388
2F_09.02.Zefiro-	SW traction block ATP	L	2F_09.02.Zefiro-Europe.CONCEPT.392

Europe.CONCEPT.572			
2F_09.02.Zefiro-Europe.CONCEPT.573	SW traction block DIS	L	2F_09.02.Zefiro-Europe.CONCEPT.394
2F_09.02.Zefiro-Europe.CONCEPT.571	CETD		2F_09.02.Zefiro-Europe.CONCEPT.396
2F_09.02.Zefiro-Europe.CONCEPT.629	Driver vigilance feedback		

Outputs:

Nr	Functional name	Type	Linked to requirement
2F_09.02.Zefiro-Europe.CONCEPT.31	--Driver Vigilance Command		

4.2 Bus Interface(base configuration)

Supplier (ATP) to TCMS SW:

Nr	Functional name	Type	Linked to requirement
2F_09.02.Zefiro-Europe.CONCEPT.351	Service brake command (SSBcomfreser)	L	2F_09.02.Zefiro-Europe.CONCEPT.331
2F_09.02.Zefiro-Europe.CONCEPT.574	Main Switch command driving (SSBpillR)	L	2F_09.02.Zefiro-Europe.CONCEPT.435
2F_09.02.Zefiro-Europe.CONCEPT.575	Raising pantograph command (SSBcompanto)	L	2F_09.02.Zefiro-Europe.CONCEPT.576
2F_09.02.Zefiro-Europe.CONCEPT.577	Change power command (SSBcambcaten)	A	2F_09.02.Zefiro-Europe.CONCEPT.457

TCMS SW to Supplier(ATP):

Nr	Functional name	Type	Linked to requirement
2F_09.02.Zefiro-Europe.CONCEPT.368	Forward (flglevainvert)	A	2F_09.02.Zefiro-Europe.CONCEPT.318
2F_09.02.Zefiro-Europe.CONCEPT.367	Backward (flglevainvert)	A	2F_09.02.Zefiro-Europe.CONCEPT.319
2F_09.02.Zefiro-Europe.CONCEPT.578	Pantograph status(flgTrolley)	A	2F_09.02.Zefiro-Europe.CONCEPT.451

TCMS SW to Supplier (DIS):
According to RD 8

4.3 I/O Interface (corridors configuration)

4.3.1 Supplier system

Input (ATP):

Nr	Functional name	Type	Linked to requirement
2F_09.02.Zefiro-Europe.CONCEPT.504	Forward	L	2F_09.02.Zefiro-Europe.CONCEPT.320
2F_09.02.Zefiro-Europe.CONCEPT.	Backward	L	2F_09.02.Zefiro-Europe.CONCEPT.321

505			
2F_09.02.Zefiro-Europe.CONCEPT.501	CabActive	L	2F_09.02.Zefiro-Europe.CONCEPT.325
2F_09.02.Zefiro-Europe.CONCEPT.502	CabNotActive	L	2F_09.02.Zefiro-Europe.CONCEPT.115

Input (DVD):

Nr	Functional name	Type	Linked to requirement
2F_09.02.Zefiro-Europe.CONCEPT.500	DvdIsolationSwitch	L	2F_09.02.Zefiro-Europe.CONCEPT.310

Outputs (DVD):

Nr	Functional name	Type	Linked to requirement
2F_09.02.Zefiro-Europe.CONCEPT.507	DvdLamp	L	2F_09.02.Zefiro-Europe.CONCEPT.198
2F_09.02.Zefiro-Europe.CONCEPT.508	DvdBuzzer	L	2F_09.02.Zefiro-Europe.CONCEPT.199

4.3.2 TCMS SW

Inputs:

Nr	Functional name	Type	Linked to requirement
2F_09.02.Zefiro-Europe.CONCEPT.512	DvdIsolationSwitch	L	2F_09.02.Zefiro-Europe.CONCEPT.311

Outputs:

Nr	Functional name	Type	Linked to requirement
2F_09.02.Zefiro-Europe.CONCEPT.514	--		

4.4 Bus Interface(base configuration)

Supplier(ATP) to TCMS SW:

Nr	Functional name	Type	Linked to requirement
2F_09.02.Zefiro-Europe.CONCEPT.517	SW traction block	L	2F_09.02.Zefiro-Europe.CONCEPT.333

TCMS SW to Supplier(ATP):

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TCMS SW to Supplier(DVD):

Nr	Functional name	Type	Linked to requirement
2F_09.02.Zefiro-Europe.CONCEPT.607	DvdTestCommand	L	2F_09.02.Zefiro-Europe.CONCEPT.316
2F_09.02.Zefiro-Europe.CONCEPT.608	DvdIsolation	L	2F_09.02.Zefiro-Europe.CONCEPT.313

5 References

RD	Code	Title	Source	Revision
RD 1	B60A.0100016.L03.00EN	ERTMS system – wiring layout and instructions for specific application on the V300 Zefiro	ASTS	Rev 03.00
RD 2	TS 373994	SISTEMA INFORMATIVO DI CONDOTTA ELENCO METADATI	TI	
RD 3	90000020.S14.EN	ATP system interface specification – ETCS/SCMT platform	ASTS	Rev 00.00
RD 4				
RD 5	3EST000225-7910	BRAKE SFRS	BT	latest
RD 6	B60A.000012.404.03I	PROCEDURE DI CAMBIO FASE E CAMBIO TENSIONE		
RD 7	4pd-pa000013-e	Cab radio functional diagram	SelexElsag	3.0
RD 8	FP02P011037B	SISTEMA INFORMATIVO DI CONDOTTA: ELENCO METADATI	AB	latest
RD 9	EC13P011957B	DIS - DESCRIPTION OF DATA RECEIVED FROM THE TRAIN	AB	latest
RD 10	EC01P014399B	Sistema vigilante - Definizione segnale di reiterazione gestito da TCMS	AB	Rev 00

Traceability matrix between SFRS and RD3

RD 3 §	Signal name	SFRS requirement	Digital/MVB	Reference to RD1	Notes
	Safe input				

2.1.6.1	Pressure Switch	N/A	Digital	N/A	Internal to ATP system
2.1.6.2	Pneumatic Plate enabled	N/A	Digital	N/A	
2.1.6.3	Cabin Selection	2F_09.02.Zefiro-Europe.TRS.124	Digital	Sheet 52 J1/B1-B2 J3/F4-F5	
2.1.6.4	Electro-Pneumatic Brake Presence	N/A	N/A	N/A	Not used by ATP application
2.1.6.5	Train integrity	N/A	N/A	N/A	Not used by ATP application
	Safe output				
2.1.7.1	Emergency Brake command	N/A	Digital	N/A	Internal to ATP system
	Non safe Inputs				
2.1.8.1	Direction controller position	2F_09.02.Zefiro-Europe.TRS.158	MVB	N/A	
2.1.8.2	Passengers Brake Inhibition Status	N/A	MVB	N/A	See ICD variable flgntalrpass
2.1.8.3	Pantograph status	2F_09.02.Zefiro-Europe.CONCEPT.451	MVB	N/A	
2.1.8.4	Doors control	N/A	MVB	N/A	Not used See ICD variable flgblocporte
2.1.8.5	Tilt control	N/A	MVB	N/A	Not used See ICD variable flgtilting
2.1.8.6	Vigilance's Pedal	2F_09.02.Zefiro-Europe.TRS.158	Digital		
2.1.8.7	Vigilance's Button	2F_09.02.Zefiro-Europe.TRS.158	Digital		
2.1.8.8	Vigilance's Exclusion	N/A	Digital	N/A	Internal to ATP system. Refer to SFRS §**Errore. L'origine riferimento non è stata trovata.**~~**Errore. L'origine riferimento non è stata trovata.**~~ for details
2.1.8.9	Sleeping input (local engine status)	N/A	N/A	N/A	Not used for ATP with pneumatic plates. See ICD variable flgstatoloco
	Non safe Outputs				
2.1.9.1	Traction cut-off command	2F_09.02.Zefiro-Europe.TRS.136	Digital	Sheet 52 J3/C1-C2	
2.1.9.2	*Electric brake command*	2F_09.02.Zefiro-Europe.TRS.132	Digital	Sheet 52 J3/J1-J2	
2.1.9.3	Pipe recharge inhibition	2F_09.02.Zefiro-Europe.TRS.170	Digital	Sheet 52 J3/K1-K2	
2.1.9.4	Vigilance's Alarm	2F_09.02.Zefiro-Europe.TRS.147	Digital	Sheet 52	

			MVB	J3/K4-K5	
2.1.9.5	Zero Speed	N/A	N/A		Not used
2.1.9.6	MCB command	2F_09.02.Zefiro-Europe.CONCEPT.435	MVB		
2.1.9.7	MCB parachute	2F_09.02.Zefiro-Europe.CONCEPT.433	Digital	Sheet 52 J1/A5-A4	
2.1.9.8	Passenger Brake inhibition	N/A	MVB		See ICD variable SSBfrenpass
2.1.9.9	Service Brake command	2F_09.02.Zefiro-Europe.TRS.132	MVB		
2.1.9.10	Inhibition of regenerative brake	N/A	MVB		See ICD variable SSBcomfrerec
2.1.9.11	Pantograph command	2F_09.02.Zefiro-Europe.CONCEPT.576	MVB		
I/O Variable included in ICD but not listed in RD3					
N/A	Open air conditioning intake	TBD	MVB		See ICD variable SSBcombocaria
N/A	Service brake status	NA	MVB		Not used See ICD variable flgfrenser
N/A	Right doors command	NA	MVB		Not used See ICD variable flgcoprtdx
N/A	Left doors command	NA	MVB		Not used See ICD variable flgcoprtsx
N/A	Redundant Vehicle's logic status	NA	MVB		Not used See ICD variable flgmasterflag

Traceability matrix between SFRS and RD1

RD 3 §	Signal name CEA	SFRS requirement	Digital/MVB	Reference to RD1 SHEET 52	Notes For speed visualization and traction enabling